

How to treat autism in children?

Evidence searched on 2026-10-07 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports **structured exercise interventions (e.g., dance, occupational therapy-based water programs) to improve motor skills** in children with autism, with dose-response effects reported for overall and gross motor function. **Feeding difficulties can be addressed with a structured, individualized treatment algorithm**, though this evidence is indirect (not specific to autism). **No direct evidence was found for social communication or emotion regulation interventions**, and no efficacy data were posted for ongoing trials.

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills. [1]

Immediate context Structured physical interventions (e.g., dance, water-based therapy) show direct evidence for motor skill improvement, but **no direct evidence supports interventions for social communication or emotion regulation**. Feeding difficulties are addressed by a structured algorithm, though this is indirect evidence (not autism-specific). No efficacy results are available for ongoing trials.

Top treatment drugs (FDA-labelled for this condition)

Drug	Indication named in the FDA label	FDA label
Aripiprazole	"Irritability Associated with Autistic Disorder"	DailyMed
Risperidone	"Treatment of irritability associated with autistic disorder"	DailyMed

1. Treatment approaches and strength of evidence

Approach	Evidence level	What the evidence shows	Reference
Exercise interventions (e.g., dance, structured walking)	Evidence-supported but limited	Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills [1].	[1]
Modified ketogenic diet + rehabilitation training	Evidence-supported but limited	The modified ketogenic diet combined with rehabilitation training was associated with greater short-term reductions in ABC and CARS scores (autism severity measures) compared to rehabilitation alone [2].	[2]
Caregiver-assisted group-based Naturalistic Developmental Behavioral Intervention (NDBI)	Emerging	Caregiver-assisted group-based NDBI showed promising improvements in developmental and social outcomes in autistic preschoolers, though nonverbal development improvements were limited to one-on-one interventions [3].	[3]
Extended reality (XR) interventions (e.g., virtual reality)	Emerging	XR interventions, particularly virtual reality, show potential for supporting children with autism, though the review highlights a lack of standardized outcomes or long-term efficacy data [4].	[4]

Approach	Evidence level	What the evidence shows	Reference
Behavioral sleep interventions	Evidence-supported but limited	Behavioral interventions consistently improve sleep outcomes in autistic children, though the review notes variability in study designs and outcomes [5].	[5]
Deep brain stimulation (DBS) for severe pediatric autism	Experimental or preliminary	A 12-month case study reported sustained reductions in core ASD symptoms (e.g., ATEC, RBS-R scores) with bilateral dual-target DBS, though this was a single-case study with no adverse events [6].	[6]
Feeding difficulties treatment (FEEDSETUP)	Indirect evidence (different or related population)	FEEDSETUP offers a structured, evidence-based approach to pediatric feeding difficulties, though it was not specifically validated for ASD [7].	[7]

Key studies

1. **Wang et al. (2026)**, Meta-Analysis: Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified [1].
2. **Liu et al. (2026)**, Randomized Controlled Trial: The modified ketogenic diet combined with rehabilitation training showed greater short-term reductions in autism severity scores (ABC, CARS) than rehabilitation alone [2].
3. **Seng et al. (2026)**, Randomized Controlled Trial: Caregiver-assisted group-based NDBI improved developmental and social outcomes in autistic preschoolers, though nonverbal gains were limited to one-on-one interventions [3].
4. **Shi et al. (2026)**, Case Reports: Bilateral dual-target deep brain stimulation sustained reductions in core ASD symptoms (e.g., ATEC, RBS-R) over 12 months in a single pediatric case [6].
5. **Mazurek et al. (2026)**, Systematic Review: Behavioral sleep interventions consistently improved sleep outcomes in autistic children, though study designs varied [5].

Standard of care: what guideline organisations say

- **American Academy of Pediatrics, 2025**: “On March 10, 2026, the FDA said there is insufficient evidence to support using leucovorin for autism and only approved it for CFD..” ([Frequently Asked Questions \(FAQs\) for Pediatricians and other Prescribing Pediatric Clinicians: Leucovorin Use in Autism and Cerebral](#))
- **US National Institute of Child Health and Human Development, 2008**: “Speech-language therapy activities can also include ways to improve social skills and social behaviors.” ([Speech-Language Therapy for Autism | NICHD - Eunice Kennedy Shriver National Institute of Child Health and Human Development](#))
- **American Occupational Therapy Association**: “It includes a description of the occupational therapy process for this population, defining the domain and describing interventions that occur within acceptable practice.” ([Occupational Therapy Practice Guidelines for Individuals With Autism Spectrum Disorder](#))
- **American Academy of Child and Adolescent Psychiatry, 2025**: “Providers typically recommend that families of children with autism seek applied behavioral analysis (ABA) for their child to improve social interaction, communication, and adaptive skills.” ([Policy Statement on Expanding Access to Care for the Autism Community](#))
- **US National Institute of Child Health and Human Development, 2004**: “In most states, each child is entitled to these services from age 3 years through high school, or until age 21, whichever comes first.” ([Educational and School-Based Therapies for Autism | NICHD - Eunice Kennedy Shriver National Institute of Child Health and Human Development](#))

One sentence from each retrieved guideline page, in the page's own words; read the page for the full recommendation.

2. Treatment Drug Landscape

2.1 FDA-approved for this condition

Drug	Approved use for this condition (FDA label wording)	Label date	FDA label
Aripiprazole	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder"	2026-03-05	DailyMed
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	2026-06-25	DailyMed

Approval applies only to the use quoted above, not to other symptoms of the condition. Doses and routes are in the linked label (not extracted here).

2.2 Other drugs described in the retrieved reviews and guidelines as used in practice

Not FDA-labelled for this condition; they are usually used for specific symptoms or co-occurring conditions, so read the source for the target symptom. Doses are not extracted.

Drug	Used for (as the source states)	Population	Source
atomoxetine	ADHD-like symptoms	not stated in this sentence	Recent Advances in Pharmacological Management of Autism: A Narrative Review
clonidine	ADHD-like symptoms	not stated in this sentence	Recent Advances in Pharmacological Management of Autism: A Narrative Review
guanfacine	ADHD-like symptoms	not stated in this sentence	Recent Advances in Pharmacological Management of Autism: A Narrative Review
methylphenidate	ADHD-like symptoms	not stated in this sentence	Recent Advances in Pharmacological Management of Autism: A Narrative Review
melatonin	sleep disturbances	not stated in this sentence	Recent Advances in Pharmacological Management of Autism: A Narrative Review

2.3 Drugs being tested in clinical trials (not established treatments)

From the trial registry. A registry entry does not show that a drug works or is used in practice.

Drug	Trial	Phase and status	Registered condition(s)	Ages enrolled	Primary outcome
buspirone	NCT07439042	Phase 4, Recruiting	Anxiety; Autism Spectrum Disorder	from 7 Years, up to 17 Years, child	Mean 16-Week Change in Pediatric Anxiety Rating Scale (PARS) 5-Item Total Score
NT1164	NCT07257939	Phase 3, Not yet recruiting	Autism; Autism Spectrum Disorder; Autism Disorder	from 6 Years, up to 21 Years, child/adult	Vineland Adaptive Behaviour Scales, 3rd Edition (Vineland-3)
KarXT	NCT07285798	Phase 3, Not yet recruiting	Irritability Associated With Autism Spectrum Disorder	from 5 Years, up to 17 Years, child	Change From Baseline in the Aberrant Behavior Checklist Irritability (ABC-I) Score at Week 8
KarX-EC	NCT07285798	Phase 3, Not yet recruiting	Irritability Associated With Autism Spectrum Disorder	from 5 Years, up to 17 Years, child	Change From Baseline in the Aberrant Behavior Checklist Irritability (ABC-I) Score at Week 8

Drug	Trial	Phase and status	Registered condition(s)	Ages enrolled	Primary outcome
Lumateperone	NCT06229210	Phase 3, Recruiting	Schizophrenia; Bipolar Disorder; Autism Spectrum Disorder	from 5 Years, up to 17 Years, child	Incidence of Common Adverse Events
Sertraline	NCT03279471	Phase 2, Completed	Autism Spectrum Disorder; Anxiety	from 8 Years, up to 14 Years, child	Change in Pediatric Anxiety Rating Scale

2.4 Recently approved or updated (FDA, EMA and CDSCO announcements)

Date	Type	Announcement	What the regulator's page says
date not stated	Regulatory announcement	FDA Takes Action to Make a Treatment Available for Autism Symptoms / FDA	"The FDA is collaborating with GSK to broaden the existing Wellcovorin label," said George Tidmarsh, M.D., Ph.D., Director of the FDA's Center for Drug Evaluation and Research. Food and Drug Administration today initiated the approval of leucovorin calcium tablets for patients with cerebral folate deficiency (CFD), a neurological condition that affects folate (a vitamin essential for brain health) transport into the brain.
date not stated	Regulatory announcement	[[PDF] guideline-clinical-development-medicinal-products-treatment-autism ...](https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-clinical-development-medicinal-products-treatment-autism-spectrum-disorder-asd_en.pdf)	No sentence stating the action was retrieved; open the page

Page wording from the regulator's own site; read the page for the full announcement.

2.5 Safety alerts (FDA boxed warnings of the drugs listed above)

Drug(s) (FDA label)	Boxed warning, in the label's words
atomoxetine	"SUICIDAL IDEATION IN CHILDREN AND ADOLESCENTS Atomoxetine increased the risk of suicidal ideation in short-term studies in children or adolescents with Attention-Deficit/Hyperactivity Disorder (ADHD). Anyone considering the use of atomoxetine in a child or adolescent must balance this risk with the clinical need. Co-morbidities occurring with ADHD may be associated with an increase in the risk of suicidal ideation and/or behavior."
methylphenidate	"ABUSE, MISUSE, AND ADDICTION Methylphenidate hydrochloride extended-release has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
Lumateperone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS; and SUICIDAL THOUGHTS AND BEHAVIORS Increased Mortality in Elderly Patients with Dementia-Related Psychosis Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. CAPLYTA is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Sertraline	"Suicidality and Antidepressant Drugs Antidepressants increased the risk compared to placebo of suicidal thinking and behavior (suicidality) in children, adolescents, and young adults in short-term studies of major depressive disorder (MDD) and other psychiatric disorders. Anyone considering the use of Sertraline hydrochloride or any other antidepressant in a child, adolescent, or young adult must balance this risk with the clinical need."
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone tablets are not approved for the treatment of patients with dementia-related psychosis."

These are the labels' own boxed warnings, not a complete safety review: read the full label.

3. Clinical trials (registry data)

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT05307744	PS128; Placebo	Phase not applicable	Terminated (2026-10-06)	Terminated early (During COVID-19, study execution); results posted (not retrieved here).
NCT06659328	Structured Walking Intervention for Children with Autism Spectrum Disorder.	Phase not applicable	Completed (2026-10-06)	Completed; results not posted/retrieved.
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-09-22)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04630847	Fecal Microbiota Transplant	Phase 1	Not yet recruiting (2026-09-18)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07439042	bupirone; Placebo	Phase 4	Recruiting (2026-09-17)	Under investigation; results not yet available; effectiveness cannot be inferred.

2 further registry record(s) about assessment, characterisation or peripheral research were retrieved and are not shown (they are not treatment trials).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

4. Evidence gaps and limitations

Conflicting or negative evidence

None was retrieved.

What remains under investigation

- **Efficacy:** The long-term effects of extended reality (XR) interventions (e.g., virtual reality) for autism symptoms remain unclear, as current evidence is limited to short-term or exploratory studies [4].
- **Feasibility:** The feasibility of caregiver-assisted group-based NDBI in resource-limited settings requires further investigation, particularly for nonverbal developmental outcomes [3].
- **Association:** The neurophysiological markers (e.g., sleep spindle characteristics) in ASD are under investigation as potential biomarkers, though their clinical utility is not yet established [8].
- **Effectiveness:** The effectiveness of structured walking interventions (NCT06659328) for gait parameters in ASD is pending, as results have not yet been posted.
- **Associations:** The role of gut microbiota (e.g., *Lactobacillus plantarum* PS128) in improving social responsiveness in ASD was investigated in a terminated trial (NCT05307744), with results posted but no clear conclusions due to operational challenges.

Limitations of this report

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses are not extracted in this report: see the linked FDA labels. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 1 retrieved record(s) were excluded because they were not about the question's subject.

- The PubMed / Europe PMC search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

References

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. [PMID 42758334](#). DOI 10.1007/s00431-026-07421-4.
2. Liu L, Zhao Q, Zhou J, et al. Evaluating the therapeutic efficacy of a modified ketogenic diet in children with autism spectrum disorder: a randomized controlled trial. *BMC pediatrics*. 2026-09-11. [PMID 42763398](#). DOI 10.1186/s12887-026-07708-3.
3. Seng GJ, Lin JY, Huang WL, et al. Feasibility and Preliminary Efficacy of a Caregiver-Assisted Group-Based Naturalistic Developmental Behavioral Intervention Program for Autistic Preschoolers: A Pilot Randomized Controlled Trial. *Autism : the international journal of research and practice*. 2026-07-23. [PMID 42494078](#). DOI 10.1177/13623613261462574.
4. Hao D, Mansi H, Elkefi S, et al. Extended reality (XR) interventions in supporting children with autism: A systematic review. *International journal of medical informatics*. 2026 Dec. [PMID 42784985](#). DOI 10.1016/j.ijmedinf.2026.106719.
5. Mazurek MO, Smith JV, Kiernan B, et al. Behavioral interventions for sleep problems in autistic children: A systematic review. *Sleep medicine*. 2026 Dec. [PMID 42685447](#). DOI 10.1016/j.sleep.2026.109242.
6. Shi M, Hu J, Pei J, et al. Dual-target deep brain stimulation for severe pediatric autism: Sustained core symptom reduction in a 12-month case study. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42759099](#). DOI 10.1016/j.clinph.2026.2112403.
7. Fisberg M, Nogueira-de-Almeida CA, Konstantyner T, et al. FEEDSETUP: feeding difficulties spectrum evaluation and treatment up-to-date proposal. *Jornal de pediatria*. 2026-09-18. [PMID 42716484](#). DOI 10.1016/j.jped.2026.101610.
8. Erçelebi H, Özbudak P, Menderes D, et al. Altered sleep spindle characteristics in children with autism spectrum Disorder: A potential neurophysiological marker. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42753649](#). DOI 10.1016/j.clinph.2026.2112405.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- [Recent Advances in Pharmacological Management of Autism: A Narrative Review](#)
- [FDA Takes Action to Make a Treatment Available for Autism Symptoms | FDA](#)
- [\[\[PDF\] guideline-clinical-development-medicinal-products-treatment-autism ...\]\(https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-clinical-development-medicinal-products-treatment-autism-spectrum-disorder-a_sd_en.pdf\)](#)
- [FDA drug label: Aripiprazole \(label effective 2026-03-05\)](#)
- [FDA drug label: Risperidone \(label effective 2026-06-25\)](#)
- [American Academy of Pediatrics: Frequently Asked Questions \(FAQs\) for Pediatricians and other Prescribing Pediatric Clinicians: Leucovorin Use in Autism and Cerebral Folate Deficiency](#)
- [US National Institute of Child Health and Human Development: Speech-Language Therapy for Autism | NICHD - Eunice Kennedy Shriver National Institute of Child Health and Human Development](#)
- [American Occupational Therapy Association: Occupational Therapy Practice Guidelines for Individuals With Autism Spectrum Disorder](#)
- [American Academy of Child and Adolescent Psychiatry: Policy Statement on Expanding Access to Care for the Autism Community](#)
- [US National Institute of Child Health and Human Development: Educational and School-Based Therapies for Autism | NICHD - Eunice Kennedy Shriver National Institute of Child Health and Human Development](#)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.